Partial-result packet for arXiv:2407.13344

Explicit affine reduction for all binary Hilbert-sphere formulas

Status. **substantial_partial_result_likely_valid**. This packet addresses Question 27.9 of Itaï Ben Yaacov, Tomás Ibarlucía, and Todor Tsankov, *Extremal models and direct integrals in affine logic*, arXiv:2407.13344v2 [1].

Source question. After proving abstractly that the theories HS_d of unit spheres of real d-dimensional Hilbert spaces have affine reduction, the source says that it does not know a syntactic proof and asks on page 121:

> Give an explicit way of approximating continuous logic formulas by affine formulas modulo HS_d.

The introduction singles out the formula $\langle x, y\rangle^3$ as an instance for which no affine approximants were known. The source page containing the formal question is reproduced below.

as desired.

Thus S belongs to the subgroup $\mathrm{Aut}(\mu) \leq \mathrm{Aut}(L^1(\Omega, M))$, and we have that

$$U = TS \in L(\Omega, \mathrm{Aut}(M)) \rtimes \mathrm{Aut}(\mu),$$

concluding the proof. □

We can now apply Theorem 22.5 and Theorem 15.6 to obtain the following.

Theorem 27.8. *The continuous logic theories* HS_d *for* $d \in \{1, 2, \ldots, \infty\}$ *have affine reduction, and the corresponding affine theories* AHS_d *are Bauer.*

The models of AHS_d *are direct integrals of spheres of d-dimensional Hilbert spaces.*

We do not know a syntactic proof of this affine reduction result.

Question 27.9. Give an explicit way of approximating continuous logic formulas by affine formulas modulo HS_d.

Question 27.10. Is AHS a Bauer theory?

28. Measure-preserving systems

Measure algebras equipped with a countable group of automorphisms are some of the most interesting examples we have for affine theories. This setup has already been considered in continuous logic: for a single aperiodic automorphism in [BBHU08], for free actions of amenable groups by Berenstein and Hen-

Scope of the result. We give an explicit uniform approximation for $\langle x, y\rangle^3$, then for every continuous function of $\langle x, y\rangle$, and consequently for every continuous logic formula with at most two free variables modulo any fixed HS_d. Question 27.9 for arbitrary finite tuples remains open.

Definitions. Affine formulas allow real affine combinations of atomic formulas and the quantifiers sup and inf. In the Hilbert-sphere language, $\langle x, z\rangle$ is atomic, and all quantified variables range over the unit sphere. Thus, for fixed real scalars a_i, the expression

$$\sup_z \sum_i a_i \langle x_i, z\rangle = \left\| \sum_i a_i x_i \right\|$$

is an affine formula modulo every HS_d.

Proof intuition. The affine fragment cannot multiply formula values, but its supremum quantifier can take a Hilbert norm. Carefully choosing two coefficients turns that norm into the square-root ridge function $t \mapsto \sqrt{1+rt}$, where $t = \langle x, y\rangle$. Finite differences in the external scalar parameter r recover every power t^k because the Taylor coefficients of $\sqrt{1+rt}$ at $r = 0$ are all nonzero. Polynomial approximation then gives every continuous function of one inner product. Finally, a binary formula on a fixed Hilbert sphere is invariant under the orthogonal group, so it is precisely a continuous function of the pair's inner product.

For three or more free variables the orbit invariant is an entire Gram matrix. The argument below supplies nonlinear functions of a single Gram coordinate, but not products of independent coordinates; this is the precise remaining obstruction to the full source question.

Proposition 1 (An affine square-root family)**.** *For $-1 < r < 1$, set*

$$a_r = \frac{\sqrt{1+r}+\sqrt{1-r}}{2}, \qquad b_r = \frac{\sqrt{1+r}-\sqrt{1-r}}{2}$$

and define the affine formula

$$N_r(x, y) = \sup_z \big(a_r\langle x, z\rangle + b_r\langle y, z\rangle\big).$$

Then, modulo every HS_d*,*

$$N_r(x, y) = \sqrt{1 + r\langle x, y\rangle}.$$

Proof. The chosen coefficients satisfy $a_r^2 + b_r^2 = 1$ and $2a_r b_r = r$. Writing $t = \langle x, y\rangle$ and using that x, y are unit vectors gives

$$N_r(x, y) = \|a_r x + b_r y\| = \sqrt{a_r^2 + b_r^2 + 2a_r b_r t} = \sqrt{1+rt}.$$

The first equality is exactly the dual formula for the norm in a real Hilbert space and remains valid if the displayed vector vanishes. □

Theorem 2 (Explicit cubic approximants)**.** *For $0 < r < 1$, the affine formula*

$$C_r(x, y) = \frac{8}{r^3}\left(N_r(x, y) - N_{-r}(x, y) - r\langle x, y\rangle\right)$$

converges uniformly, modulo every HS_d*, to $\langle x, y\rangle^3$ as $r \downarrow 0$. The same formula and error bound work in every dimension.*

Proof. For $t \in [-1, 1]$, Proposition 1 gives

$$C_r(t) = \frac{8}{r^3}\left(\sqrt{1+rt} - \sqrt{1-rt} - rt\right).$$

Taylor's theorem for $s \mapsto \sqrt{1+s}$, uniformly on $[-1/2, 1/2]$, yields

$$\sqrt{1+s} - \sqrt{1-s} - s = \frac{s^3}{8} + O(s^5).$$

Taking $s = rt$ shows

$$\sup_{|t|\le 1} |C_r(t) - t^3| = O(r^2),$$

which proves uniform convergence. □

Proposition 3 (Every one-variable connective)**.** *Let $f \in C([-1,1])$ and let $\varepsilon > 0$. There is an explicitly constructible affine formula $A_{f,\varepsilon}(x,y)$ such that, in every real Hilbert sphere and uniformly in x, y,*

$$|A_{f,\varepsilon}(x,y) - f(\langle x,y\rangle)| < \varepsilon.$$

Proof. We first recover every monomial. Fix $k \geq 1$ and, for $0 < h < 1/k$, let

$$P_{k,h}(x,y) = \frac{1}{k!\binom{1/2}{k}h^k}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}N_{jh}(x,y).$$

This is an affine formula. For $t = \langle x,y\rangle$, let $g_t(r) = \sqrt{1+rt}$. The numerator above is the kth forward difference $\Delta_h^k g_t(0)$. The integral formula for finite differences gives

$$\frac{\Delta_h^k g_t(0)}{h^k} = \int_{[0,1]^k} g_t^{(k)}(h(s_1 + \cdots + s_k))\, ds_1 \cdots ds_k.$$

The derivatives are jointly continuous for $|t| \leq 1$ and $|r| < 1$. Consequently the right-hand side converges uniformly in t to

$$g_t^{(k)}(0) = k!\binom{1/2}{k}t^k.$$

Thus $P_{k,h} \to \langle x,y\rangle^k$ uniformly as $h \downarrow 0$.

Now choose, for example, a Bernstein polynomial p satisfying $\|f - p\|_\infty < \varepsilon/2$. Write $p(t) = \sum_{k=0}^{m} c_k t^k$. Replace each t^k by P_{k,h_k}, choosing h_k sufficiently small that the sum of the resulting errors is less than $\varepsilon/2$; use the constant affine formula 1 for $k = 0$. The resulting affine combination is $A_{f,\varepsilon}$. □

Theorem 4 (Binary affine reduction)**.** *Fix $d \in \{1, 2, \ldots, \infty\}$. Every continuous logic formula modulo HS_d having at most two free variables can be uniformly approximated by affine formulas. For $d \geq 2$, the construction is the one in Proposition 3 applied to the continuous function on $[-1,1]$ induced by the formula.*

Proof. Let $\varphi(x,y)$ be a continuous logic formula. In a Hilbert sphere of fixed dimension $d \geq 2$, two ordered pairs of unit vectors are in the same orthogonal-group orbit exactly when they have the same inner product. (For $d = \infty$, this is equivalently the standard description of binary types in the complete theory of infinite-dimensional real Hilbert spaces.) Formula interpretations are automorphism invariant. Hence there is a function $f_d : [-1,1] \to \mathbb{R}$ such that

$$\varphi(x,y) = f_d(\langle x,y\rangle) \pmod{\mathrm{HS}_d}.$$

The function f_d is continuous: one may evaluate φ on the continuous family

$$x = e_1, \qquad y_t = te_1 + \sqrt{1-t^2}\, e_2, \quad -1 \leq t \leq 1.$$

Proposition 3 now gives the required affine approximants. If $d = 1$, the only possible inner products are -1 and 1, so every binary invariant is exactly an affine function of $\langle x,y\rangle$. Unary formulas are constant by transitivity of the orthogonal group, and sentences are constant modulo the complete theory HS_d. □

Verification notes. The proof uses only affine operations: each N_r is one supremum quantifier applied to an affine combination of atomic inner products, while all finite differences and polynomial combinations use real linear combinations. There is no use of a nonlinear connective inside an approximating formula.

The included script `code/check_approximants.py` uses 80-digit arithmetic and samples 2001 equally spaced values of $t \in [-1, 1]$. For the cubic approximants it observes the predicted quadratic error decay as r is halved; it also checks the forward-difference formulas for powers 1 through 6. These numerical checks are sanity checks only; uniform convergence is proved above.

Limitations and novelty check. This is a partial answer to Question 27.9: arbitrary-arity formulas remain untreated. The source's highlighted cubic instance is completely resolved, as is the entire binary fragment. The construction is uniform over Hilbert dimension once the target binary connective has been chosen, although the function f_d associated with a general formula may depend on d.

Before promotion, the lightweight run indexes were searched for arXiv:2407.13344, "affine reduction", "Hilbert sphere", and the cubic formula; no duplicate packet was found. The current arXiv record is v2 and still contains Question 27.9. Bounded searches of arXiv for the exact question, the paper title plus later work, and explicit affine Hilbert approximants found no later stated syntactic solution. This check is not exhaustive, so novelty remains subject to expert review.

Human-review recommendation. Review as a substantial partial solution. The central points to check are the coefficient identity in Proposition 1, the uniform finite-difference limit in Proposition 3, and the claim that binary types of a fixed Hilbert-sphere theory are parametrized by the single inner product. The arbitrary-arity gap should remain explicit.

References

[1] I. Ben Yaacov, T. Ibarlucía, and T. Tsankov, *Extremal models and direct integrals in affine logic*, arXiv:2407.13344v2, 2024.